RESEARCH | Article in Press

Seq2DomML: Protein domain boundary prediction from bacterial protein sequences using a machine learning framework

Alana Monks · Azam Asilian Bidgoli · Michael D. L. Suits

Received: 14 July 2026 / Accepted: 15 September 2026
© The Author(s) 2026

Abstract

Purpose The identification of protein domain boundaries is important for designing stable single-domain constructs for protein structural solution. For proteins without experimentally determined structures, boundary selection often relies on conserved domain annotations, predicted structures, or iterative truncation screening. Seq2DomML is developed as a sequence-based machine learning framework for predicting residue-level domain partitions in bacterial proteins without requiring structural coordinates or homologous domain assignments as input.

Methods Structural domain annotations from the Evolutionary Classification of Domains (ECOD) database were converted into binary residue labels, including residues within annotated domains and domain boundaries, linkers, and terminal flanking regions. Each residue was initially encoded using 2389 amino acid identity, positional, physicochemical, and interaction features. The feature space was reduced through correlation-based pruning and multi-objective evolutionary feature selection, and the selected features were used to train a bidirectional long short-term memory (LSTM) model.

Results Seq2DomML was evaluated on a held-out test set after sequences sharing at least 35% global sequence identity with the training set were removed and was compared with established tools, including InterPro, Pfam, and NCBI CDD. The classification metrics showed that Seq2DomML outperformed the comparison tools, achieving a macro F1-score of 0.734 and a within-domain F1-score of 0.924. Its domain count mean absolute error was 0.952, the second-lowest value after Pfam. Seq2DomML additionally achieved the highest mean domain Intersection over Union (IoU) of 0.846, indicating the strongest spatial agreement with the ECOD-derived domain regions among the evaluated methods.

Conclusion Seq2DomML demonstrates that sequence-derived residue features can support prediction of structurally defined domain partitions without direct structural input. The model provides a complementary approach to conserved domain annotation tools and may help prioritize candidate splice regions for construct design, particularly when structural or homologous evidence is limited.

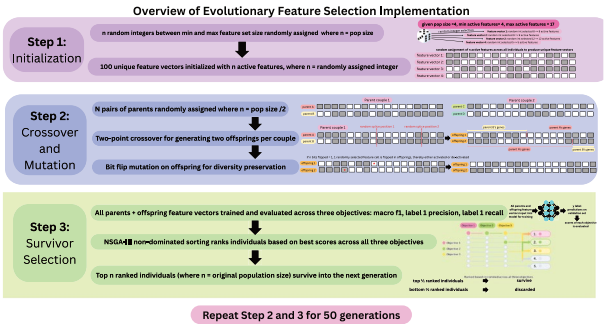
This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.



https://doi.org/10.1186/s12859-026-06666-y

Published online: 06 October 2026

Graphical abstract



Keywords Protein construct design · Protein domain annotation · Protein domain prediction · Long short-term memory model · Feature selection · Machine learning

Introduction

Protein domains are independently folded structural units that play fundamental roles in protein function and organization. Accurate identification of domain boundaries is therefore important for structural biology, protein annotation, and protein engineering. In structural determination workflows, identifying appropriate domain boundaries is also a critical step in protein construct design. Beyond structural biology, reliable domain boundary prediction also facilitates the functional interpretation of proteins, comparative genomics, and protein engineering by identifying independently folded functional units that can be studied or manipulated individually.

Protein structure determination depends not only on the use of an appropriate experimental method but also on the design of a suitable protein construct for expression and solubility. Typically, the most experimentally manageable construct is not the full-length protein but a stable region that preserves an independently folded domain unit while excluding flexible linkers, disordered termini, or neighboring domains that may introduce conformational heterogeneity [1]. Identifying appropriate domain boundaries is therefore a central step in construct design, yet it remains difficult when no experimentally determined structure is available. Furthermore, domain boundaries cannot always be inferred from local sequence composition alone because domain formation is influenced by long-range structural interactions and evolutionary constraints. Accurate sequence-based prediction of domain boundaries therefore remains challenging, particularly for large multi-domain proteins or proteins lacking experimentally determined structures [2].

Experimental workflows therefore often require the creation and screening of many truncation constructs to identify positions that preserve the structure and function of a protein domain and yield a successful crystallization hit [3]. The primary drawback of this approach is that screening numerous truncated constructs increases the time and cost of structure-determination workflows. Computational tools that predict candidate domain boundaries can help reduce this burden.

Existing sequence-based domain annotation resources, such as Pfam [4] and InterPro [5], identify conserved domains and protein families primarily through sequence homology and curated domain models. Although highly effective for well-characterized proteins, these resources may fail to recognize domains with few well-annotated homologs [5]. Consequently, methods are needed that can predict domain boundaries directly from amino acid sequences without relying on structurally characterized homologs. Such methods are particularly valuable for recently discovered proteins, poorly characterized protein families, and proteins whose domain organization is not represented in existing annotation databases.

To address this challenge, this paper proposes Seq2DomML, a sequence-based machine learning (ML) framework for residue-level protein domain boundary prediction. Rather than transferring domain annotations from

homologous proteins, Seq2DomML learns domain partitioning from structurally characterized bacterial proteins and predicts domain boundaries using only sequence-derived information. The framework is designed to reduce the trial and error involved in truncation screening by helping prioritize splice sites that preserve independently folded domains while excluding flexible or heterogeneous regions [6]. Decades of structural biology research have shown that multi-domain proteins, proteins with extensive loop regions, and proteins with unconstrained linkers are more difficult to crystallize. Motivated by this challenge, Seq2DomML provides a sequence-based framework for predicting residue-level domain boundaries that can guide construct design, particularly for proteins lacking experimentally determined structures or well-annotated homologous domains.

The proposed framework uses structural domain annotations from the Evolutionary Classification of Domains (ECOD) database [7] to generate binary residue-level labels for a dataset of structurally characterized bacterial protein sequences from the Protein Data Bank (PDB) [8]. Following target labeling, thousands of sequence-derived physicochemical features from the GenomeNet database [9] are extracted for each residue and then reduced to an optimized subset through two-stage feature selection. The final ML model is trained using a bidirectional long short-term memory (LSTM) network, enabling residue-level domain boundary predictions for sequences that lack structural characterization or well-annotated homologous domains.

By facilitating the identification of stable, independently folded domains for protein construct design, Seq2DomML aims to address a difficult component of structure-determination workflows. This capability is particularly valuable for proteins lacking experimentally determined structures or reliable homologous domain annotations, thereby extending structural analysis beyond well-studied proteins.

Background and literature review

Protein domains

In the context of this model, a protein domain is defined as a contiguous sequence region corresponding to an independently folded structural unit. Protein domains are commonly understood as modular units of protein structure and function that may fold with some degree of independence from the rest of a polypeptide chain [10]. Large proteins are often composed of multiple domains connected by intervening regions defined as linkers [11]. Since the positive class does not represent a domain itself, but rather, the residues belonging to sequence contexts surrounding or separating domains, this class semantically covers linkers, domain boundaries, and terminal flanking regions. This distinction is important because the model is not simply predicting whole domain membership, but identifying the sequence positions most relevant to domain partitioning.

Flanking regions in this study are defined as terminal sequence segments located outside the first or last annotated domains of a protein chain. These regions may include disordered termini, affinity tag adjacent sequences, or flexible extensions that are not part of the independently folded domain units being targeted for structural solution. In construct design, such regions are often removed because terminal flexibility and disorder can reduce sample homogeneity and therefore interfere with stable packing interactions [12].

The inclusion of flexible linkers in protein constructs can increase conformational heterogeneity because the connected regions can adopt multiple relative orientations [13]. This variability can hinder protein packing into crystal lattices and ultimately reduce diffraction quality.

Flexible loops and linkers may also lack packing constraints because they sample many backbone torsions, enabling broad conformational variability that makes rigid crystal packing more difficult to achieve [14]. In turn, removing linkers from protein samples can promote crystal formation by reducing conformational heterogeneity and enhancing crystal lattice formation [12].

Multi-domain proteins often adopt multiple inter-domain arrangements [12], increasing conformational heterogeneity and disrupting the formation of well-ordered crystal lattices. They also make it more difficult to identify experimental conditions that support successful crystallization. In particular, multi-domain proteins may

be difficult to stabilize under a single set of solution conditions because the pH, salt concentration, or additives that preserve the stability and homogeneity of one region may not be optimal for another. These differences can increase partial unfolding and aggregation while reducing the number of productive crystallization hits [15].

Because of these effects, constructs containing flexible linkers, flanking regions, or multiple domains may be undesirable when well-ordered crystal packing is required. Crystallography workflows therefore commonly pursue domain isolation, as truncating multi-domain proteins into individual functional units can improve crystallization success [16].

Long short term memory (LSTM) models in protein sequence prediction tasks

The selected architecture for the workflow is a bidirectional long short-term memory (LSTM) model [17], chosen for its ability to model both local residue neighborhoods and long-range relationships along a sequence.

LSTMs are recurrent neural networks designed for sequential data, in which information from earlier positions can influence later predictions. LSTM models maintain an internal memory state that is updated as the sequence is processed and controlled through three learned gating mechanisms: the input, forget, and output gates [18]. This architecture was originally developed to mitigate the vanishing-gradient problem observed in conventional recurrent neural networks [17], thereby improving their applicability to time-series and other sequential datasets.

Bidirectional LSTM variants are increasingly used in proteomics-based prediction tasks because they integrate context from both the N- to C-terminal and C- to N-terminal directions [19]. This capability is especially valuable for tasks such as the one considered in this study, in which a residue label depends on surrounding sequence patterns at multiple scales. Several successful applications motivated the selection of an LSTM architecture for Seq2DomML. For example, CLPred uses an LSTM to predict whether k-mers from a protein sequence will yield diffraction-quality crystals [20]. Other applications of LSTM architectures include protein function prediction [21] and protein-protein interaction binding-site prediction [22].

Overall, LSTMs naturally support residue-level sequence-labeling tasks such as domain boundary identification, in which an output label is produced at each sequence position rather than a single label per protein. This architecture is therefore suitable for Seq2DomML because the prediction at a given residue depends on both its local neighborhood and distal sequence context.

It is nevertheless important to consider this choice in relation to other widely used architectures for protein representation, including attention-based mechanisms and pretrained protein language models for sequence-based prediction. Seq2DomML did not adopt these architectures because it was specifically designed as a task-specific supervised predictor that operates directly on explicitly engineered residue-level sequence features rather than as a large pretrained representation model. A bidirectional LSTM provided a targeted architecture for learning contextual dependencies across the residue sequence while retaining a relatively compact model formulation. Recent protein-modeling studies have also combined pretrained protein-language representations with BiLSTM layers to model sequential and long-range residue dependencies further [23]. Accordingly, the choice of a BiLSTM in Seq2DomML should not be interpreted as a claim that recurrent architectures are universally superior to attention-based or protein-language-model approaches, but rather as a task-specific choice aligned with the residue-level sequential formulation of the problem.

Additionally, the computational and memory requirements of standard self-attention mechanisms scale quadratically with sequence length, which would increase the computational burden of repeatedly training candidate models during evolutionary feature selection [24]. The BiLSTM was therefore selected as a practical sequence-labeling architecture that can model bidirectional residue dependencies while remaining compatible with the repeated model training required by the proposed feature-selection procedure.

Multi-objective feature selection

Multi-objective feature selection refers to an optimization problem where the goal is to select an optimal set of features to improve performance of a machine learning task (e.g., classification) across multiple objectives that are often simultaneously conflicting. Rather than performing optimization to improve a single score, multi-objective optimization ranks or preserves candidate solutions according to tradeoffs across objectives. A solution is considered non-dominant when no other solution performs better across all objectives and strictly better in at least one objective, allowing the algorithm to approximate the Pareto front for tradeoff solutions [25].

To solve such optimization problems, evolutionary algorithms, inspired by biological evolution, are a highly effective class of optimization algorithms [26]. During the optimization, a population of candidate solutions is iteratively improved through selection, crossover, and mutation operators. Candidate solutions are evaluated according to their fitness, and stronger performing solutions are preferentially carried forward as surviving populations across generations [25].

Non-Dominated Sorting Genetic Algorithm (NSGA-III) is an evolutionary algorithm designed for multi-objective optimization problems, where the goal is to improve performance across multiple objectives that are often simultaneously conflicting. Rather than performing optimization to improve a single score, NSGA methods rank candidate solutions according to Pareto dominance. A solution is considered non-dominant when no other solution performs better across all objectives, allowing the algorithm to approximate the Pareto front for tradeoff solutions.

NSGA-III is a widely used implementation of this framework that uses reference points distributed across the objective space to preserve diversity among candidate solutions. Candidate solutions are sorted into non-dominated fronts, and the top-ranking fronts are preferentially carried forward as surviving populations across generations. The reference-point-based selection mechanism allows NSGA-III to maintain a diverse set of candidate feature subsets across competing objectives [27], motivating its selection for this study.

The Non-Dominated Sorting Genetic Algorithm (NSGA-III) [27] is a many-objective evolutionary algorithm designed to solve optimization problems involving multiple conflicting objectives. It first ranks candidate solutions using non-dominated sorting, where a solution is considered non-dominated if no other solution is at least as good in all objectives and strictly better in at least one objective. The set of non-dominated solutions forms an approximation of the Pareto front, representing different trade-offs among the objectives. To maintain diversity, NSGA-III associates solutions with a predefined set of reference points and selects individuals to achieve a well-distributed approximation of the Pareto front. This reference-point-based selection enables NSGA-III to maintain both convergence and diversity more effectively than traditional crowding-distance-based approaches, particularly in many-objective optimization problems.

Existing sequence-based domain prediction approaches

ECOD is considered a gold-standard resource because it uses the atomic coordinates of experimentally determined structures, together with homology information, to classify domains based on structural and evolutionary relationships. Other tools commonly used to annotate domains in structure-determination workflows include Pfam [4], InterPro [5], and NCBI CDD [28]. Unlike ECOD, these tools do not require structural input, making them applicable to structurally uncharacterized sequences.

Pfam is a database of protein families/domains in which each entry is represented by a seed multiple sequence alignment and a profile hidden Markov model (profile HMM). Pfam is still a sequence-based annotation method at query time, but modern Pfam curation increasingly uses AlphaFold predictions [29] as structural evidence to refine families and improve domain boundary predictions [4]. The 2025 Pfam update reports that Pfam release 37.0 contains 21,979 families and 709 clans, with 76.3% sequence coverage and 48.8% residue coverage across the UniProtKB [30] reference proteomes.

NCBI CDD is a collection of multiple-sequence-alignment-derived domain and protein-family models represented as position-specific scoring matrices (PSSMs) [28]. NCBI CDD is a valuable comparator because it performs sequence-to-domain annotation without requiring a query structure, although it remains a library-search system based on curated domain models. Version 3.20 contains 59,693 protein and protein-domain models, including 19,178 Pfam-derived models, 18,882 in-house-curated CDD models, and 4541 superfamily clusters, as well as 42,937 site annotations across 15,943 staff-curated models [28].

InterPro is an integrative protein sequence classification resource. It does not rely on one modeling strategy; instead it integrates signatures from multiple member databases, including Pfam [4], CDD [28], SMART [31], PANTHER [32], PROSITE [33] and others. InterPro consolidates overlapping evidence from multiple databases into a single annotation framework, attaches InterPro entries, abstracts, and often GO terms, and now provides representative domains chosen to maximize coverage while minimizing overlap [5]. InterPro is best thought of as a meta-resource with its domain annotation framework and is not exclusively a single homogeneous predictor. Its output quality depends on the underlying member database signatures.

ECOD is widely used as a domain classification resource built around proteins with experimentally determined 3D structures, providing domain assignments and boundary definitions on PDB chains [7]. It is best described as the structural reference standard used to derive domain annotations, because it classifies domains using a combination of sequence and structural data, and its hierarchy is built to emphasize evolutionary relationships, including distant homology, rather than only local sequence similarity or shared topology [7]. Although, because ECOD is fundamentally anchored to existing structural coordinates, it is limited in its utility for unsolved protein sequences.

Recent work using the Domain Parser for AlphaFold Models (DPAM) [34] has sought to address this gap by combining predicted AlphaFold structures with sequence information to produce domain assignments. However, the confidence of these assignments is limited when the underlying AlphaFold structure predictions have low confidence. Across 48 proteomes, DPAM assigned approximately 90% of residues to an annotation category [35], although a substantial proportion could not be mapped straightforwardly to ECOD domains. Approximately 66% of residues were confidently assigned to an ECOD homologous group, 4% were classified as globular but could not be confidently assigned to ECOD, 12% were classified as flexible linker or disordered regions, and another 12% occurred in low-confidence AlphaFold regions that were not assigned as domains.

Although Pfam, InterPro, and NCBI CDD provide valuable domain annotations, their performance is inherently tied to the availability and quality of homologous sequence evidence. For proteins from poorly characterized families, or for sequences with few well studied homologs, these tools may return incomplete annotations, overly broad domain matches, missing boundaries, or no confident domain assignment at all [36], which limits the success of domain isolation in crystallography, Cryo-EM, and Nuclear Magnetic Resonance (NMR) towards already-well studied proteins. The homology reliance of these tools therefore presents a strength for domain isolation when there are sufficient homologs, yet remains weaker for poorly studied protein families posing constraints in the structural solution process for each approach.

A key difficulty in X-ray diffraction is that splice boundaries must be chosen to retain a complete, independently folding unit [10]. When boundaries cut into structured elements or disrupt domain integrity, the resulting construct can misfold, lose function, or become unstable. This process can therefore be high stakes as a single position can separate crystallizable from non crystallizable variants, or can be the difference between a functional domain unit, as opposed to a dysfunctional domain unit from a partial domain construct [3]. Moreover, domains can also be made up of non-contiguous sequence elements, meaning that these domains would be defined from multiple sequence segments. For example, in Cryo-EM, flexible regions and mobile domains often blur density, reduce local resolution, and make classification and refinement harder. Sequence guided removal of such flexible regions or separation of domain units would therefore improve particle homogeneity, and increase local map interpretability [37]. For NMR based structural studies, careful construct optimization is often necessary to improve solubility, folding behavior, and spectral quality. By suggesting domain consistent splice regions from

sequences alone, an annotation tool may help prioritize constructs that better preserve independently folded units while excluding destabilizing flanking segments [38].

In contrast to these annotation tools, Seq2DomML does not rely on domain conservation, sequence alignment, experimentally determined structures, or computationally predicted structures; it uses only an amino acid sequence as input. The trained model learns sequence-based signals associated with domain and non-domain residues, allowing predictions to be made directly from amino acid sequence features. Seq2DomML is not intended to replace curated domain databases but to provide an additional prediction layer when homologous domain evidence is limited, structural coordinates are unavailable, or predicted structures are too uncertain to support confident domain annotations.

Methods

The overall aim of this workflow is to develop a sequence-based model that predicts residue-level domain labels directly from amino acid sequence features and provides domain-partitioning guidance for structurally uncharacterized sequences. As summarized in Fig. 1, the workflow consists of three main stages.

First, bacterial protein chains solved by X-ray crystallography were collected from the PDB, filtered, deduplicated, and matched to ECOD annotations. These annotations were then converted into binary residue labels, with residues within annotated domains encoded as domain residues (label 0) and residues outside annotated domains encoded as non-domain residues (label 1). Next, feature extraction was performed to derive 2389 features for each residue in each sequence. These features included amino acid identities, positional descriptors, amino acid indices, and interaction-matrix features. Finally, the feature space was reduced through correlation-based pruning and subsequently optimized using multi-objective evolutionary feature selection. The resulting candidate feature subsets were used to train bidirectional LSTM models with class- and residue-specific loss weighting. Model configurations were evaluated on the validation set to select the final feature subset and prediction threshold. The selected model was then applied to the held-out test sequences, followed by post-prediction smoothing of non-domain regions and evaluation using residue-level classification, domain-overlap, and implied domain-count metrics. Finally, Seq2DomML predictions were compared with those of Pfam, InterPro, and NCBI CDD after

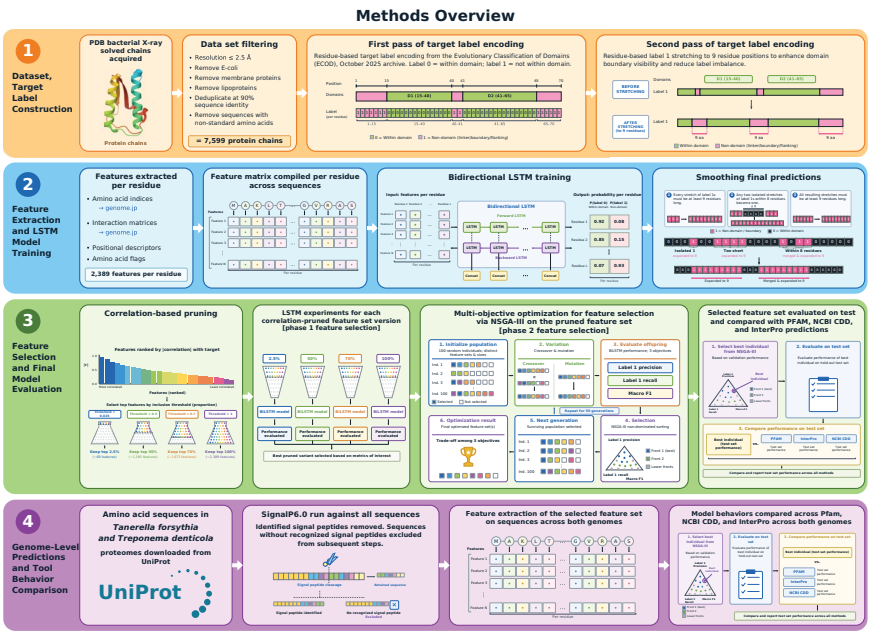


Fig. 1 Overview of the Seq2DomML methodological workflow. The workflow consists of three main stages. First, bacterial protein chains are collected, filtered, and converted into residue-level labels using ECOD annotations. Second, sequence-derived features are extracted for each residue and used to train a bidirectional LSTM model. Finally, the feature space is optimized through correlation-based pruning and evolutionary feature selection, and the resulting model is evaluated against Pfam, NCBI CDD, and InterPro.

all outputs were converted to the same residue-level label representation, providing a common framework for evaluating domain-partitioning performance.

The following sections describe each methodological phase in greater detail, including dataset construction, target label encoding, feature extraction, model training, feature reduction, evolutionary feature selection, and the final model evaluation.

Dataset construction and target label encoding

Dataset construction began by acquiring protein chains from the PDB [8] that were taxonomically assigned to the kingdom Bacteria, had a refinement resolution of ≤ 2.5 Å, and had been solved by X-ray diffraction. Proteins from *Escherichia coli* were removed because the abundance of this species could introduce species-specific bias into the training and evaluation datasets. To reduce heterogeneity further, membrane proteins were removed because of the distinct characteristics of their domains and the specialized approaches required for their structural determination [39]. Lipoproteins were also excluded because bacterial lipoproteins are often embedded in or closely associated with membranes [40].

After the initial preprocessing stage, 45,680 protein chains remained. The October 2025 ECOD annotation archive was then used to obtain target labels and covered 43,278 sequences in the dataset. Sequences containing non-standard amino acids, such as X, were subsequently removed to enable amino acid feature extraction.

Following these filtering steps, sequences that were near-duplicates of one another were removed to preserve dataset diversity and reduce bias. This step is important in protein-based machine learning because data leakage can occur when highly similar protein chains or mutant variants appear across the training and evaluation sets, producing overly optimistic estimates of generalization [41].

This step is performed by removing sequences using 90% global sequence identity clustering. Firstly, candidate pairs were pre-filtered by a sequence length difference of less than 10 residues and greater than 90% identity across the first 50 amino acids, before being globally aligned using Biopython"s PairwiseAligner tool. A 90% sequence-identity threshold has also been used in protein sequence-based machine learning studies as a practical redundancy-reduction strategy for removing duplicate or near-duplicate protein sequences while preserving sufficient dataset size for model development. For example, PSO-LocBact removed proteins exceeding 90% sequence identity using CD-HIT before randomly partitioning the resulting nonredundant dataset into training and test sets, with the authors noting that the higher threshold was selected to preserve adequate sample representation during training for underrepresented classes [42]. Despite the effectiveness of this initial de-duplication step for reducing near-duplicate sequences across the complete dataset, an additional cross-split sequence-similarity filtering step was performed before final test-set evaluation using a stricter sequence-identity cutoff. This training-to-test homology filtering step is described later in the Methods section. Table 1 provides a summary of the data acquisition and filtering process.

After preprocessing, the ECOD annotations for the 7599 protein chains were converted into binary residue-level labels that distinguish within-domain from non-domain residues across each protein sequence. The within-domain class (the negative label) includes all residues within an annotated domain, whereas the non-domain class (the positive label) includes all residues outside an annotated domain, including domain boundaries, linkers, and flanking segments. Because the two classes were imbalanced, an initial expansion step extended domain boundaries to a minimum length of nine residues. This step increased the model"s exposure to the minority class and ensured that flanking and linker regions were not underrepresented. It also reflects the biological interpretation of domain boundaries as transition regions rather than precisely defined single-residue positions. Furthermore, because protein construct design typically tolerates small shifts in splice-site selection around domain boundaries, representing these regions as short contiguous segments better aligns the training labels with their intended downstream application. Figure 2 provides an overview of this process.

Table 1 Construction and filtering of the bacterial protein sequence dataset

Step	Inclusion / Exclusion Criteria	Remaining Sequence Chains
Initial bacterial PDB X-ray chains	Resolution≤ 2.5Å	71,942
Removal of *Escherichia coli* sequences	Taxonomic classification is not *Escherichia coli*	60,791
Removal of membrane and lipoproteins	Does not contain ``lipoprotein"", ``lipo"", or ``membrane""	45,680
ECOD annotation matched	Domain annotation present in ECOD archive from October 2025	43,278
Deduplication at 90% global sequence identity	Retain one representative from each similarity cluster	7599

Summary of the sequential inclusion and exclusion steps used to construct the final dataset from bacterial protein chains solved by PDB X-ray crystallography

The first column identifies each stage of dataset preparation, the second specifies the criterion applied at that stage, and the third reports the number of sequence chains remaining after the criterion was applied

Chains were filtered by structural resolution, taxonomic classification, protein type, availability of an ECOD domain annotation, and sequence similarity

Deduplication at 90% global sequence identity produced the final dataset of 7599 sequence chains used for target label construction and subsequent model development



Fig. 2 Example of residue label encoding and boundary expansion for PDB 1H6C chain B. The raw amino acid sequence is shown together with the initial and expanded residue labels derived from the three ECOD annotated domains. In the initial encoding, label 0 residues shown in black represent positions within annotated domains, while label 1 residues shown in pink represent terminal flanking regions, linkers, and single residue domain edges. Each continuous label 1 region is then expanded to a minimum length of nine residues, as shown in the final row, to improve representation of the minority class and increase the visibility of domain boundary regions during model training.

Finally, the constructed dataset was split into training, validation, and test sets for model development and evaluation. The training set contained 5700 samples, the validation set contained 666 samples, and the held-out test set contained 1235 samples.

Feature extraction

Using the target-labeled dataset, features were extracted to encode residue-specific signals at all sequence positions. These features included one-hot encodings of the 20 standard amino acids, relative residue positions, amino acid indices, and interaction matrices from the AAindex database [43]. In total, 2389 features were extracted for each residue.

The one-hot amino acid encodings preserved the categorical identity of each residue, allowing the model to distinguish among the 20 standard amino acids at each sequence position. However, because one-hot encoding

does not directly describe biochemical similarities or differences among amino acids, amino acid index features were also included to represent residue-level physicochemical properties such as hydropathy, charge, size, polarity, flexibility, and secondary-structure preferences. These properties are relevant to domain boundary prediction because physicochemical tendencies differ across structural contexts, particularly among stable domain interiors, flexible linkers, solvent-exposed loops, and terminal flanking regions. Positional features were included because domain boundaries and flanking regions are not distributed randomly across protein sequences, and terminal regions may exhibit different label patterns from internal domain boundaries. Interaction-matrix features were included to encode pairwise amino acid relationships, such as statistical contact and substitution potentials, allowing the model to infer how a residue relates to neighboring residues. Together, these feature categories represent each residue using both its amino acid identity and broader biochemical context. Table 2 summarizes the extracted feature types and their counts.

Model training

The LSTM was trained on the binary residue labels in the training sequences to predict a sequence of binary labels from the amino acid features at each input position. The validation set was used to select the best-performing model from the experimental trials, and the test set was held-out for final model evaluation and comparison with other tools. After the final training epoch, a smoothing function was applied before evaluation to keep the model"s non-domain residue predictions biologically plausible and consistent with the expansion rule used during target-label encoding. Predicted non-domain stretches separated by short stretches of domain residues were merged when the intervening gap was no more than nine residues. This smoothing step was not used during training; it was applied only after prediction to reduce biologically implausible fragmentation of domain-boundary predictions. The nine-residue minimum was consistent with the target-label encoding procedure, in which domain boundaries were deliberately expanded to nine residues.

Handling label imbalance through loss customizations

Despite the earlier expansion of domain boundaries to increase the representation of the minority label, the remaining class imbalance required additional treatment to improve boundary detection. To address this issue, effective class weights were incorporated into the binary cross-entropy loss during LSTM training. This approach has been shown to increase the influence of minority classes in imbalanced datasets [44]. It assigns larger weights to underrepresented classes while accounting for the diminishing contribution of repeated samples within the majority class, thereby increasing the influence of minority-class residues during optimization.

For each residue with true binary label $y_i \in \{0, 1\}$ and predicted probability $\hat{y}_i$, the weighted binary cross-entropy loss is defined as

$$\mathcal{L}_{\mathrm{BCE}} = -\alpha_{y_i}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)],$$

Table 2 Overview of the feature space used in Seq2DomML

Feature Type	Description	Count
One-hot amino acid encoding identities	One-hot encodings of each of the 20 standard amino acids	20
Positional features	Relative position across sequence, and relative distance to N-termini and C-termini	3
Amino acid indices	Physicochemical indices [43]	566
Interaction matrices	Interaction matrices, contact potentials, and substitution matrices derived from [43]	1800

Breakdown of the four feature categories used to encode each amino acid position within the input protein sequences

The first column identifies the feature category, the second summarizes the sequence or biochemical information represented by that category, and the third reports the number of individual features contributed to each residue representation

The feature space includes 20 amino acid identity indicators, three positional descriptors, 566 physicochemical amino acid indices, and 1800 interaction matrix features, producing a total of 2389 features per residue before feature reduction and selection

where α_{y_i} denotes the class-specific weight associated with the true residue label. In this study, $y_i = 0$ corresponds to domain residues, whereas $y_i = 1$ corresponds to boundary residues, including domain boundaries, linkers, and terminal flanking regions. The class-specific weight α_{y_i} was computed using the effective number of samples approach. First, the effective number of samples for each class c was calculated as

$$E_{n_c} = \frac{1 - \beta^{n_c}}{1 - \beta},$$

where n_c denotes the number of training residues belonging to class c, and β is the effective number weighting parameter. The raw class-balanced weight was then computed as $w_c = \frac{1}{E_{n_c}}$. To maintain a consistent overall loss magnitude, the class weights were normalized such that they summed to two:

$$\tilde{w}_c = \frac{2w_c}{w_0 + w_1}.$$

Finally, the residue-specific weight used in the loss was assigned according to the true residue label:

$$\alpha_{y_i} = y_i \tilde{w}_1 + (1 - y_i)\tilde{w}_0.$$

Consequently, boundary residues received the normalized weight $\tilde{w}_1$, whereas domain residues received $\tilde{w}_0$.

In addition to the effective class weights, a second weighting scheme was introduced to prevent the model from becoming overconfident in predicting non-domain residues at sequence termini. This issue arises because terminal residues are overrepresented within the non-domain class. The residue-based loss was therefore divided into three categories: (1) negative residue weights for true within-domain residues, (2) positive terminal weights for true non-domain residues in stretches adjacent to either sequence terminus, and (3) internal positive weights for true non-domain stretches not adjacent to either terminus, including domain boundaries and linkers. A residue-type weight γ_i was additionally assigned as follows:

$$\gamma_i = \begin{cases} \gamma_{\text{domain}}, & \text{if } y_i = 0, \\ \gamma_{\text{terminal}}, & \text{if } y_i = 1 \text{ and residue } i \text{ belongs to a terminal positive region}, \\ \gamma_{\text{internal}}, & \text{if } y_i = 1 \text{ and residue } i \text{ belongs to an internal positive region}. \end{cases}$$

Subsequently, the loss for each residue was obtained by multiplying the residue-type weight, the class-balancing weight, and the binary cross-entropy loss as follows:

$$\mathcal{L}_i = \gamma_i \alpha_{y_i} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)].$$

These values were aggregated across all residues in each batch to compute the loss for each epoch.

Feature selection

Feature selection has also been applied in protein-sequence classification tasks, including post-translational modification-site prediction and organelle-protein classification, to reduce high-dimensional feature representations and retain informative variables for downstream prediction [45, 46].

With 2389 features, the computational cost of the model was high, particularly when running the evolutionary feature-selection algorithm. The full feature set was also expected to contain redundancy, motivating feature reduction before final model training.

A two-phase feature-selection framework was proposed to improve computational efficiency and feature-subset quality. Because evolutionary algorithms become computationally expensive in very high-dimensional search spaces [47], the first phase removes irrelevant and redundant features through correlation-based pruning. In the

second phase, a multi-objective evolutionary algorithm identifies candidate feature subsets from the reduced search space.

Phase 1: Correlation-based feature pruning

The first phase of feature selection evaluated model performance using feature sets containing different proportions of features ranked according to their correlations with the target variable.

The point-biserial correlation coefficient was calculated to quantify the relationship between each numerical feature and the binary target variable. Absolute correlation values were used to rank the features according to the strength of their association with the target. Multiple feature subsets were then generated by retaining different proportions of the ranked features. Smaller subsets contained only the features most strongly correlated with the target, whereas larger subsets included progressively more weakly correlated features.

Finally, the LSTM model was trained on each pruned feature set to evaluate model performance and identify a promising feature pool that would reduce the complexity and improve the efficiency of the subsequent feature-selection phase.

Phase 2: Multi-objective evolutionary feature selection

Following correlation-based feature pruning, multi-objective evolutionary optimization was used in the second phase to identify improved subsets from the retained feature pool. Each individual represented a candidate subset selected from the pruned feature space. To mitigate premature convergence and preserve exploration, the worst-performing 10% of individuals, ranked by macro F1-score, were replaced every ten generations with newly initialized individuals. Feature subsets were evaluated according to the performance of a bidirectional LSTM trained using only the selected features as input. Because the goal was not only to maximize overall classification performance but also to improve detection of the minority non-domain class, the subsets were assessed using multiple validation metrics, including macro F1-score, non-domain precision, and non-domain recall. This approach preserved candidate solutions representing different trade-offs among residue-level classification metrics. Figure 3 summarizes the evolutionary optimization process.

The primary practical constraint on the use of NSGA-III was the computational cost of evaluating candidate feature subsets with an LSTM model. To reduce runtime, the optimization stage limited the maximum number of LSTM training epochs, applied early stopping based on the validation macro F1-score, and used only 15% of the training and validation sets. The purpose of this stage was not to estimate the final performance of each model but to rank feature subsets efficiently and identify promising regions of the search space. The evolutionary

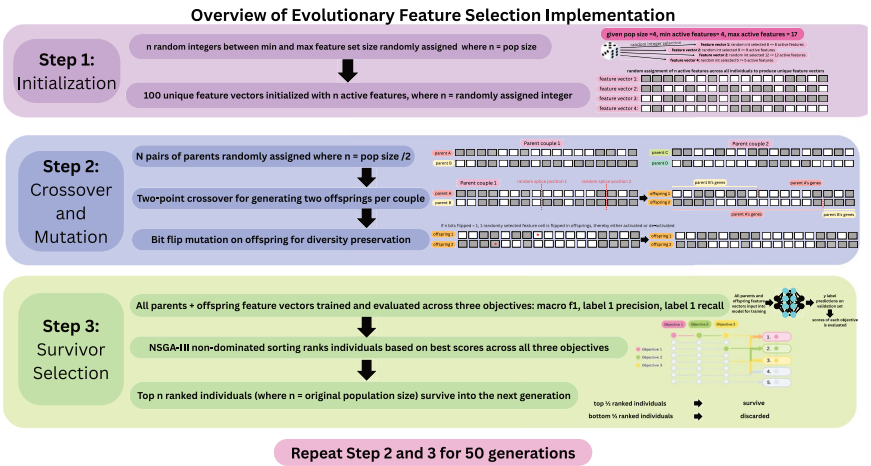


Fig. 3 Evolutionary feature selection workflow. Evolutionary feature selection in Seq2DomML is performed in three main stages. During initialization, a population of unique binary feature vectors is created. During crossover and mutation, candidate vectors are paired to generate offspring. During survivor selection, all parent and offspring feature subsets are evaluated using the LSTM model.

search therefore functioned as a screening step that narrowed the candidate subsets before more computationally intensive retraining.

This phase produced a final population of candidate feature subsets rather than a single optimized subset. Each candidate in the final population was retrained on the complete training set and evaluated on the complete validation set. The final feature subset was selected based on the optimization-objective scores and the mean absolute error of the implied number of domains predicted for each sequence.

Experiments and results

This section presents the results of the experiments described above and explains how they informed parameter setting, model tuning, and final model selection.

Evaluation metrics

Model performance was evaluated using both residue-level classification metrics and domain architecture level metrics. At the residue-level, performance was measured using domain residue F1-score, non-domain residue precision, non-domain residue recall, non-domain residue F1-score, and macro F1-score. Domain residue F1-score measures how accurately residues within annotated domains were recovered, while non-domain residue precision and recall evaluate the model"s ability to identify boundary, linker, and terminal flanking residues without over assigning these regions. Macro F1-score was included to summarize performance across both residue classes while reducing the dominance of the majority domain residue class. Since the intended application is domain partitioning for construct design, domain count mean absolute error (MAE) was also used to evaluate how closely each predicted residue label pattern resulted in the true number of implied domain units in each sequence. This metric is calculated by counting the number of domains predicted against the true number of domains determined by ECOD, thereby providing an important method of evaluating the overall domain assignment architecture implied by the model predictions. In order to additionally assess the segmentation and spatial overlap between predicted and true domain regions, domain Intersection over Union (IoU) was calculated. For each protein, domain IoU measures the overlap between residues predicted as within-domain and residues annotated as within-domain in the ECOD-derived true labels relative to the combined extent of the predicted and true domain regions. IoU ranges from 0 to 1, with higher values indicating greater spatial agreement between the predicted and true domain regions. The mean domain IoU across the evaluated protein sequences was reported for each method.

LSTM hyperparameter setting

The first set of experiments evaluated how the LSTM model responded to different loss-weighting and prediction-threshold settings before feature selection was performed. These trials were designed to identify a configuration that could preserve strong classification of domain residues while improving detection of the minority non-domain class, which includes boundary, linker, and terminal flanking residues. Specifically, the experiments varied the effective class weight beta term used, the residue type weights assigned to domain residues, terminal non-domain residues, and internal non-domain residues, as well as the final prediction threshold used to convert residue probabilities into binary labels. All results in this subsection are reported on the validation set. For the residue type weights and beta terms, a wide combination of configurations were screened in order to find a parameter set that met a high standard for this model.

Overall, the beta term had a clear effect on the trade-off between non-domain recall and precision across the hyperparameter trials. Using a more aggressive beta value of 0.999 improved non-domain recall, although at the cost of substantially lower precision. Specifically, the trial with the highest non-domain recall achieved a recall of 0.964 but a precision of only 0.174. In contrast, a beta value of 0.99999 yielded the strongest macro F1-scores

and substantially higher non-domain precision. The trial with the highest non-domain precision reached 0.953 but achieved a recall of only 0.365. These results show that the highest-recall configurations were associated with severe precision loss, whereas the highest-precision configurations had reduced non-domain recall. The balanced trials in Table 3 show that both beta values could produce useful precision-recall compromises, although the more aggressive beta value generally appeared in models that maintained recall with a moderate reduction in precision.

The residue-type weight configurations also affected whether the model behaved as a recall-oriented, precision-oriented, or more balanced predictor. Overall, downweighting the positive terminal weights while upweighting the internal positive weights yielded the highest non-domain recall, particularly for domain boundaries. The highest-recall models used low terminal positive weights of 0.10 and higher internal positive weights of either 0.70 or 1.00. However, higher negative residue weights, such as NRW = 1.00 or 0.75, combined with higher terminal and internal positive weights, shifted the model toward stronger non-domain precision but weaker non-domain recall. The balanced results in Table 3 suggest that no single residue-type weight configuration was universally optimal. However, downweighting the terminal positive weight contributed most consistently to maintaining non-domain recall. This observation is consistent with the imbalance among non-domain residue types: terminal residues are more abundant than internal boundary residues. Without appropriate weighting, an LSTM may therefore rely excessively on sequence position and assign the non-domain class primarily near sequence termini rather than considering the physicochemical context of residues near internal domain boundaries.

The prediction threshold further controlled the final precision-recall balance. The highest non-domain recall values occurred at lower prediction thresholds of 0.45 to 0.55, where recall ranged from 0.911 to 0.964 but precision remained low, ranging from 0.174 to 0.192. In contrast, the highest non-domain precision values occurred at thresholds of 0.675 to 0.700, where precision exceeded 0.944 but recall decreased to approximately 0.365--0.382. The balanced results in Table 3 fall between these extremes, with prediction thresholds ranging from 0.45 to 0.625. Thus, moderate thresholds were most effective in preserving non-domain recall while avoiding the severe precision loss observed in the highest-recall trials. These results led to the selection of a beta value of 0.99999, a negative residue weight of 0.5, a terminal positive weight of 0.25, and an internal positive weight of 1.0 for the subsequent correlation-based feature-pruning trials. The complete selected LSTM training configuration is presented in Table 4.

Correlation-based feature pruning experiments

The next experiment evaluated model performance using different proportions of features ranked by their correlation with the target variable. The point-biserial correlation coefficient was calculated for every feature, and each resulting subset retained a distinct proportion of the *n* most highly correlated features.

Table 3 LSTM hyperparameter trials with balanced non-domain precision and recall

NRW	TPW	IPW	Beta	Pred. Thresh.	Macro F1	ND Prec.	ND Recall	WD F1	ND F1
1.000	0.500	1.500	0.99999	0.450	0.756	0.693	0.467	0.932	0.580
1.000	1.000	1.000	0.999999	0.500	0.764	0.721	0.466	0.935	0.594
0.500	0.250	1.000	0.99999	0.550	0.749	0.675	0.459	0.930	0.567
1.000	0.800	1.200	0.999999	0.550	0.767	0.738	0.459	0.936	0.598
0.750	0.750	1.500	0.999999	0.625	0.760	0.715	0.458	0.934	0.586

Five validation trials selected to represent balanced performance for the non-domain class"s precision and recall

Trials were included when non-domain precision exceeded 0.650 and non-domain recall exceeded 0.450

NRW denotes the residue weight applied to within-domain residues, TPW denotes the weight applied to terminal non-domain residues, and IPW denotes the weight applied to internal non-domain residues associated with boundaries and linker regions

The column abbreviations depict specific metrics as follows: *ND* non-domain, *WD* within-domain, *Pred. Thresh.* prediction threshold, *Prec.* precision

Table 4 Final LSTM training configuration

Parameter	Value
Learning rate	1×10^{-4}
Spatial Dropout 1D	0.1
LSTM dropout	0.1
Loss function	Binary cross entropy
Optimizer	Adam
Bidirectional	True
Early stopping	No Validation macro F1-score improvement for 10 epochs
Batch size	50
LSTM units	100
CB beta	0.99999
Negative residue weight (NRW)	0.5
Terminal positive weight (TPW)	0.25
Internal positive weight (IPW)	1.0
Prediction threshold	0.55

The table summarizes the final hyperparameter and loss-weighting configuration used to train the bidirectional LSTM model after validation-based screening of learning settings, class-weighting parameters, residue-type weights, and prediction thresholds

NRW denotes the weight applied to within-domain residues, TPW denotes the weight applied to terminal non-domain residues, and IPW denotes the weight applied to internal non-domain residues associated with boundary and linker regions

Table 5 Validation performance across correlation-based feature inclusion thresholds

Threshold	# Features	Macro F1	WD F1	ND F1	ND Prec.	ND Rec.
0.025	59	0.758	0.935	0.582	0.739	0.425
0.050	119	0.766	0.937	0.596	0.769	0.423
0.100	238	0.776	0.939	0.612	0.805	0.419
0.150	358	0.786	0.942	0.630	0.849	0.411
0.200	477	0.780	0.941	0.620	0.825	0.415
0.300	716	0.783	0.941	0.625	0.833	0.417
0.400	955	0.787	0.942	0.632	0.850	0.415
0.500	1194	0.778	0.940	0.615	0.806	0.424
0.600	1433	0.787	0.942	0.632	0.849	0.416
0.800	1911	0.785	0.941	0.628	0.836	0.419
1.000	2389	0.792	0.943	0.641	0.875	0.407

The table compares the feature subsets produced during the first phase of feature reduction by retaining different proportions of the original 2389 residue features. The inclusion threshold indicates the proportion of the complete feature space retained, while the second column reports the corresponding number of features used as model inputs

A threshold of 1.0 represents the complete feature set, while smaller thresholds represent progressively reduced feature spaces selected according to their correlation based ranking

The column abbreviations depict specific metrics as follows: *WD* within-domain, *ND* non-domain, *Prec.* precision, *Re.* recall

As shown in Table 5, larger feature sets generally yielded higher macro F1-scores, whereas smaller feature sets yielded stronger non-domain recall. To balance these outcomes while favoring the more difficult objective of improving non-domain recall, 25% of the complete feature set was retained as the feature pool for subsequent evolutionary feature selection.

Evolutionary feature selection experiments

The evolutionary feature-space optimization was initialized with 100 individuals, each containing a random number of selected features between 40 and 597 to provide diversity in both subset size and composition. Offspring were produced using two-point crossover followed by bit-flip mutation, with the number of flips decreasing every

ten generations to promote greater diversity during the early generations. The mutation operator initially flipped five bits per individual and decreased to one bit per individual by generations 40--50. This schedule allowed the search to explore a broad range of feature combinations early in the run and shift toward smaller refinements of stronger candidate subsets in later generations. Table 6 presents the complete operator settings.

The feature-selection process showed gradual improvement across generations, particularly in macro F1-score and non-domain precision. The best macro F1-score increased early in the search and then stabilized for much of the run, suggesting that relatively strong feature subsets were identified within the first several generations. Non-domain precision showed a larger stepwise improvement, increasing substantially during the early generations and again near the end of the run. This pattern indicates that the evolutionary search identified feature combinations that reduced false-positive non-domain predictions. Non-domain recall remained comparatively stable and lower than precision, reflecting the difficulty of detecting the minority non-domain class. The primary purpose of this stage was therefore to guide the search toward promising feature subsets; the final population of 100 surviving individuals was subsequently retrained on the complete dataset.

After optimization and retraining of the final population on the complete training set, each feature subset was evaluated on the validation set at six prediction thresholds: 0.35, 0.40, 0.45, 0.50, 0.55, and 0.60. This process produced 600 candidate models. Given the notable trade-offs among the optimization objectives, the final model was selected based on its overall validation performance before evaluation on the held-out test set.

To identify candidate trials for final selection, the 600 post-feature selection trials were first filtered by applying minimum performance requirements to all three optimization objectives and the domain count MAE. Candidate trials were required to satisfy non-domain precision greater than 0.6, non-domain recall greater than 0.45, a macro F1-score greater than 0.74, and a domain count MAE less than 0.94. These thresholds were selected to exclude trials that achieved strong performance in only one metric while performing poorly in another. In application terms, the selected candidates needed to retain sufficient sensitivity to boundary, linker, and terminal residues, avoid excessive over assignment of non-domain regions, preserve strong overall residue classification, and maintain a reasonable estimate of the implied number of domain units. This filtering step yielded 16 candidate trials, each coming from distinct individuals evaluated at unique prediction thresholds, as shown in Table 7. Across all trials, visible tradeoffs between residue-level classification metrics and the range of error in the implied domain counts, which made the decision of selecting the final best individual feature set less straightforward.

As shown in Fig. 4, non-domain precision and recall displayed the expected inverse relationship, with higher precision generally corresponding to lower recall. This trade-off was important during final selection because excessive emphasis on precision could increase the overall macro F1-score while reducing the detection of true boundary and linker regions. Conversely, excessive emphasis on recall could increase the overprediction of non-domain residues. A similar trade-off was observed between non-domain recall and macro F1-score, with higher recall generally corresponding to a lower macro F1-score. Across the comparison plots, the highest classification scores did not necessarily correspond to the lowest domain count MAE.

Table 6 NSGA-III operator settings for evolutionary feature selection.

GA Operator	Value / Rule
Minimum feature set size	40
Maximum feature set size	597
Population size	100
Reference points	91
Crossover	Two point
Mutation rate	5 bit flips, decaying by 1 every 10 generations
Replacement rule	Replace the bottom 10% by macro F1 score every 10 generations with newly initialized random individuals

Summary of the evolutionary algorithm settings used during the second phase of feature selection in Seq2DomML

Candidate feature subsets contained between 40 and 597 active features and were evolved within a population of 100 individuals. NSGA-III used 91 reference points to preserve diversity across the three optimization objectives

Table 7 The 16 trials of post-feature selection individuals evaluated at different prediction thresholds meeting the criteria of macro F1-score greater than or equal to 0.74, non-domain precision greater than or equal to 0.60, non-domain recall greater than or equal to 0.45, and domain count MAE less than 0.94

# Feat.	Pred. Thresh.	WD F1	ND F1	ND Prec.	ND Recall	Macro F1	DC MAE
434	0.450	0.934	0.584	0.717	0.451	0.759	0.923
441	0.450	0.932	0.578	0.692	0.463	0.755	0.914
492	0.450	0.932	0.576	0.685	0.468	0.754	0.887
527	0.450	0.932	0.575	0.686	0.465	0.754	0.907
477	0.450	0.932	0.575	0.685	0.464	0.753	0.902
490	0.400	0.930	0.572	0.671	0.474	0.751	0.916
474	0.450	0.930	0.571	0.666	0.476	0.751	0.919
516	0.400	0.931	0.570	0.680	0.460	0.750	0.923
511	0.450	0.931	0.569	0.676	0.462	0.750	0.926
515	0.400	0.930	0.569	0.668	0.471	0.750	0.883
474	0.400	0.930	0.567	0.667	0.467	0.748	0.928
531	0.400	0.929	0.566	0.660	0.471	0.747	0.893
424	0.450	0.928	0.565	0.645	0.485	0.746	0.923
464	0.400	0.929	0.563	0.654	0.471	0.746	0.929
516	0.400	0.926	0.558	0.631	0.486	0.742	0.907
454	0.400	0.926	0.555	0.634	0.476	0.741	0.898

The column abbreviations depict specific metrics as follows: *WD* within-domain, *ND* non-domain, *Feat.* features, *Pred. Thresh.* prediction threshold, *Prec.* precision, *DC MAE* domain count mean absolute error

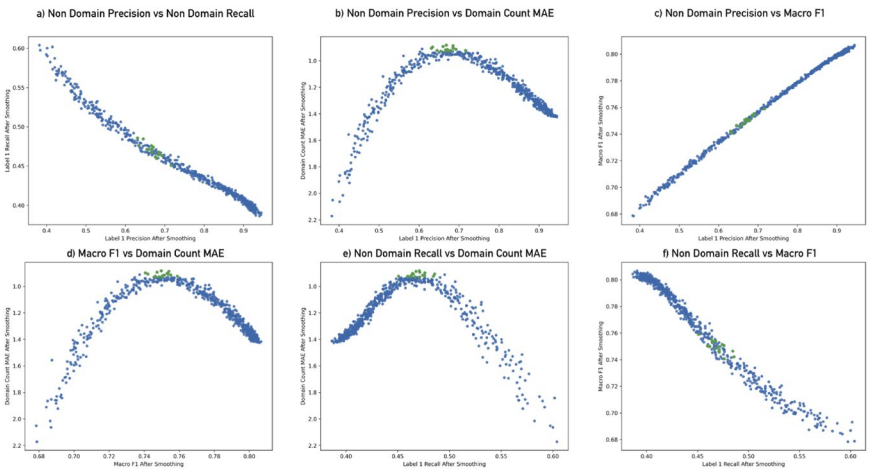


Fig. 4 Pairwise performance trade-offs across final feature selection trials. The six scatterplots compare the validation performance of the 600 trials. Each blue point represents one feature subset and prediction threshold combination after prediction smoothing, while the green points identify the 16 candidate trials retained after applying the final performance filters. Each subfigure represents the trade-off between two evaluation metrics.

The results additionally demonstrate that the macro F1-score increased strongly with non-domain precision, but the highest macro F1-scores did not correspond to the lowest domain count MAEs. In contrast, domain count MAE displayed a curved relationship with non-domain precision, non-domain recall, and macro F1-score. The lowest domain count errors occurred in an intermediate performance region rather than at the extremes of precision, recall, or macro F1-score. This finding indicates that balancing the objectives is necessary to improve the accuracy of implied domain-count predictions.

Among the 16 filtered candidate trials in Table 7, the final selected feature set contained 531 features and used a prediction threshold of 0.40. This trial was selected because it achieved one of the lowest domain count MAEs and one of the highest non-domain recalls among the filtered candidates. Non-domain recall had been the most difficult aspect of the proposed model to improve. This feature set and prediction threshold were therefore selected for final test-set evaluation and comparison with the other tools.

Comparison with state-of-the-art domain annotation tools

The final Seq2DomML model, using the selected feature set and hyperparameters described above, was compared on the test set with three established domain annotation tools: Pfam, InterPro, and NCBI CDD. Unlike Seq2DomML, these tools rely largely on homology-based approaches to domain annotation. The comparison therefore highlights the relative strengths and limitations of each method in different sequence contexts. Predictions from all methods were converted to a consistent binary residue-label format to support a common evaluation.

Prior to the final held-out test evaluation, an additional cross-split sequence-similarity filtering step was applied to provide a stricter assessment of model generalization without the influence of cross-split homology. Although the initial deduplication at 90% sequence identity reduced the presence of near-duplicate proteins in the complete dataset, proteins with lower but still substantial similarity could remain across the subsequently generated training and test partitions. Such train-test homology can lead to overly optimistic performance estimates in protein sequence-based machine learning because predictions may be made for proteins closely related to sequences encountered during model development [48]. Therefore, each protein in the held-out test set was compared with the training sequences, and test proteins sharing at least 35% global sequence identity with any training protein were excluded. Global alignments were generated using BLOSUM62 with gap penalties, and sequence identity was calculated from exact amino acid matches across each alignment. Importantly, this filtering was applied only to the held-out test set; the training set and previously selected model parameters were left unchanged. Following this procedure, the final held-out test set contained 664 protein sequences. This test set was subsequently used for the final evaluation of Seq2DomML and the corresponding comparisons with Pfam, InterPro, and NCBI CDD.

To evaluate each tool, predicted domain annotations were converted into the same residue-level binary label format used for Seq2DomML. The within-domain class represented residues within annotated domains, whereas the non-domain class represented residues outside annotated domains, including boundaries, linkers, and terminal flanking regions. The same non-domain expansion rule used during target encoding was applied to the other tools so that domain boundaries were expanded to nine residues, enabling a consistent comparison.

As shown in Table 8, Seq2DomML achieved the highest macro F1-score of 0.734 among the evaluated methods. The model also achieved the highest non-domain precision at 0.640, indicating that its non-domain predictions were less prone to false-positive assignments than those of the other tools. NCBI CDD achieved the second-highest non-domain precision at 0.517, whereas Pfam and InterPro achieved values of 0.368 and 0.339, respectively.

Seq2DomML also achieved the highest domain Intersection over Union (IoU) of 0.846, compared with 0.817 for NCBI CDD, 0.710 for Pfam, and 0.675 for InterPro. Domain IoU measures the residue-level overlap between predicted and true within-domain regions relative to their combined extent, with higher values indicating greater

Table 8 Reduced test set performance comparison across domain annotation tools

Tool	WD F1	ND Prec.	ND Recall	ND F1	Macro F1	DC MAE	D IoU
InterPro	0.814	0.339	0.722	0.461	0.638	1.047	0.675
Pfam	0.834	0.368	0.733	0.490	0.662	0.865	0.710
NCBI CDD	0.905	0.517	0.504	0.510	0.708	1.410	0.817
Seq2DomML	0.924	0.640	0.446	0.543	0.734	0.952	0.846

This table compares the performance of InterPro, Pfam, NCBI CDD, and Seq2DomML on the held-out test after the 35% similarity reduction from train-to-test

It shows the residue classification metrics, domain Intersection over Union, and domain count mean absolute error values for each tool

Higher values indicate stronger performance for each classification metric and the domain IoU metric, whereas lower values indicate stronger performance for domain count MAE

The column abbreviations depict specific metrics as follows: *WD* within-domain, *ND* non-domain, *Prec.* precision, *DC MAE* domain count mean absolute error, *D IOU* Domain Intersection Over Union

spatial agreement. The higher IoU achieved by Seq2DomML therefore indicates that its predicted domain regions showed the greatest overlap with the reference domain annotations among the evaluated methods.

The main trade-off for Seq2DomML was observed in non-domain recall. InterPro and Pfam achieved higher non-domain recall values of 0.733 and 0.722, respectively, whereas the Seq2DomML model achieved a lower value of 0.446. This finding indicates that InterPro and Pfam detected more true non-domain residues but did so with considerably lower precision, suggesting that many true domains were not detected and were therefore labeled as non-domain. This difference reflects an important limitation of the comparison: the meaning of a non-domain prediction differs between Seq2DomML and the homology-based tools Pfam, NCBI CDD, and InterPro. In Seq2DomML, a non-domain is a defined region outside a structured domain. For the other tools, however, any region in which no domain is detected is treated as non-domain. Consequently, a lack of domain recognition by these tools may inflate non-domain recall, and this metric may not fully reflect domain-segmentation accuracy.

In contrast, Seq2DomML produced fewer non-domain predictions overall, but these predictions had substantially higher precision. Thus, the model provided a more selective assignment of the non-domain class.

The residue-level heatmaps in Fig. 5 provide a visual comparison of these prediction patterns across a sample of 25 sequences. InterPro and Pfam show broader and more fragmented non-domain predictions across many sequences, consistent with their higher recall but lower precision. NCBI CDD produces more conservative domain-region predictions but visibly fails to identify a substantial proportion of non-domain regions relative to the ground-truth annotations. Seq2DomML shows a more selective non-domain prediction pattern, with fewer scattered predictions than InterPro and Pfam while still capturing boundary-associated regions. The longer non-domain stretches observed for InterPro and Pfam also reflect their homology-based focus: domains with few known homologs may remain unrecognized and consequently be represented as non-domain regions.

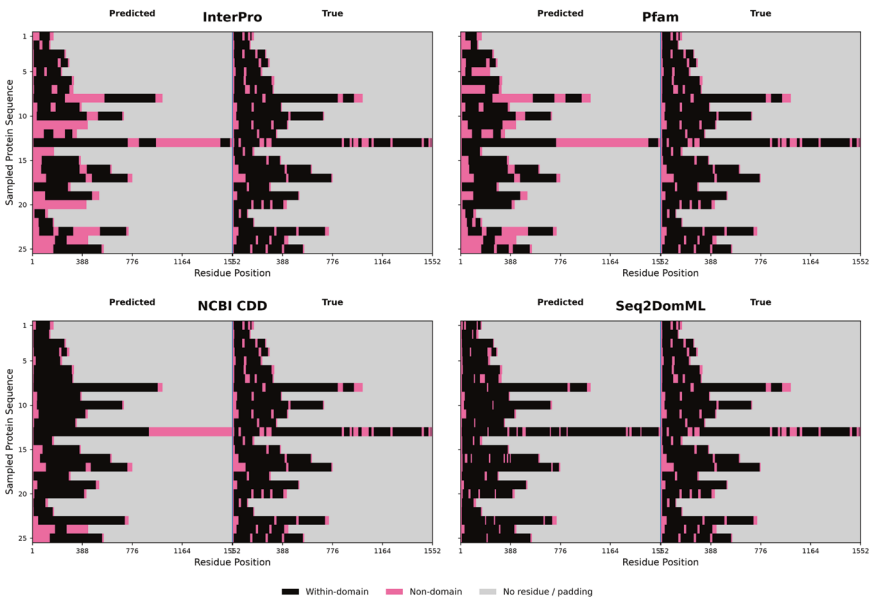


Fig. 5 Comparison heatmaps of residue-level predictions across a sample of test sequences for each domain annotation method. The heatmaps compare residue labels predicted by InterPro, Pfam, NCBI CDD, and Seq2DomML with the corresponding ECOD-derived reference labels for the same 25 randomly selected sequences from the homology-reduced test set. Each row represents one protein sequence, and the horizontal axis represents residue position. Within each subfigure, the predicted labels are shown on the left and the reference labels on the right, allowing spatial agreement between predicted and reference domain architectures to be examined across methods. Black represents within-domain residues, whereas pink represents non-domain residues, including domain boundaries, linkers, and terminal flanking regions. Grey regions indicate positions beyond the ends of shorter sequences after alignment to a common plotting width. The repeated reference-label heatmap in each subfigure provides a consistent basis for visually comparing the methods.

In addition to residue-level classification metrics, each method was evaluated according to its implied number of domains per sequence, with a domain defined as a stretch of more than ten predicted within-domain residues. This analysis was included because the intended use of the model is not only to identify individual residue labels but also to support domain-partitioning decisions for construct design. For each sequence, the predicted label pattern was converted into an implied domain count, and the mean absolute error between the predicted and reference counts was calculated.

As shown in Table 8, Pfam achieved the lowest domain count MAE at 0.865, followed by Seq2DomML at 0.952, InterPro at 1.047, and NCBI CDD at 1.410. Thus, Pfam produced the closest overall domain-count estimates, whereas Seq2DomML achieved the second-lowest error. Although NCBI CDD performed relatively well in residue-level classification, it had the highest domain count MAE, indicating that residue-level F1-scores alone do not fully capture domain-architecture performance.

The domain-count error categories in Fig. 6 provide a more detailed view of how each method produced its overall domain count MAE. Pfam produced the largest number of exact predictions, correctly estimating the domain count for 289 sequences, followed by InterPro for 243, Seq2DomML for 214, and NCBI CDD for 85. However, the broader error distributions reveal an important distinction between exact-count matching and overall prediction behavior. Pfam, InterPro, and NCBI CDD more often predicted too few domains than too many. In contrast, Seq2DomML exhibited a more varied distribution of domain-count errors and a reduced tendency toward underprediction. Although Seq2DomML overpredicted the domain count in some cases, this behavior may be useful in applications that seek to identify smaller, previously unrecognized domain units.

InterPro and Pfam also tended to under-predict the domain counts more often than they overpredicted it, although this tendency was less pronounced than for NCBI CDD. Seq2DomML showed a more balanced distribution of underprediction and overprediction errors, with 219 sequences predicted as one domain too few and 283

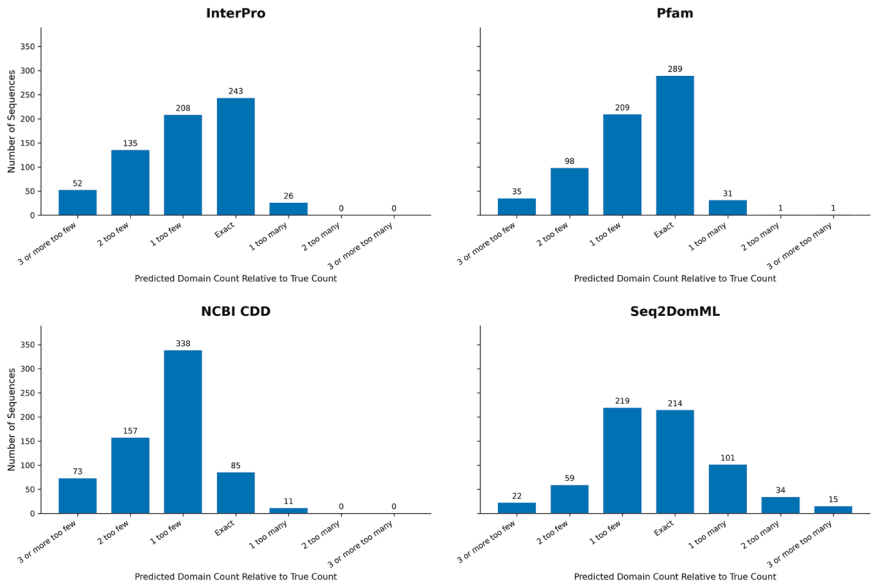


Fig. 6 Comparison of domain error category counts across annotation tools. The plots summarize the difference between the number of domain units predicted by InterPro, Pfam, NCBI CDD, and Seq2DomML and the corresponding true number derived from ECOD annotations on the homology-reduced test set. The horizontal axis groups sequences according to the predicted domain count relative to the true count, while the vertical axis shows the number of sequences assigned to each category. Exact indicates that the predicted and true domain counts were equal. The categories 1 too few, 2 too few, and 3 or more too few indicate increasing underprediction of the true number of domains, whereas 1 too many, 2 too many, and 3 or more too many indicate increasing overprediction. The number displayed above each bar gives the total number of sequences within that category. Together, the four panels show the distribution and direction of domain count errors produced by each annotation method.

predicted as one domain too many. While Pfam performed best for exact domain count recovery, Seq2DomML achieved the second lowest domain count MAE whilst avoiding the severe underprediction pattern observed for NCBI CDD, demonstrating a key strength of the proposed framework. Taken together with its strongest macro F1-score, non-domain precision, and non-domain F1-score, this indicates that the final model offers a comparatively balanced prediction profile: it remains competitive for domain count recovery while providing stronger residue-level discrimination of non-domain regions.

Beyond the residue-level metrics, predictions from each tool were visualized by mapping the assigned labels onto the corresponding experimentally determined structures. The examples in Figs. 7–9 were selected from 100 randomly sampled comparison structures to illustrate the strengths and potential applications of Seq2DomML. In these renderings, residues assigned to the non-domain class are shown in white, including linkers, terminal flanking regions, and boundary-associated residues. Structured domain regions are shown in color: red for the first domain, orange for the second, and yellow for the third. For Pfam, NCBI CDD, and InterPro, however, an extended white region does not necessarily represent a predicted non-domain region; it may instead indicate that no domain was detected in that portion of the sequence. This pattern is particularly apparent in the InterPro and Pfam outputs.

Because non-domain regions were deliberately expanded to nine residues during target-label encoding, predicted white boundary regions may extend into structured secondary elements in both the ECOD-derived reference labels and the predicted labels. Therefore, white residues within otherwise structured regions should not automatically be interpreted as non-domain; they may instead indicate the predicted location of a domain boundary.

For PDB structure 1V8C Chain D (Fig. 7), the ECOD-derived labels show two major structured domains separated by a short non-domain boundary region. Pfam and InterPro recover a similar two-domain organization, with the left beta-rich domain and the right helical domain separated by a white linker-like interval. NCBI CDD also broadly captures this organization, although its segmentation appears more conservative and less sharply separated at the boundary. Seq2DomML produces a visually comparable two-domain partition but additionally assigns a small internal region within the left domain to the non-domain class. This extra white region may represent a false-positive boundary prediction, although it occurs near an exposed loop at the domain edge rather than within the buried domain core. Some apparent false-positive boundary predictions may therefore reflect structurally plausible local flexibility or subdomain-like partitioning not explicitly represented in the ECOD annotation.

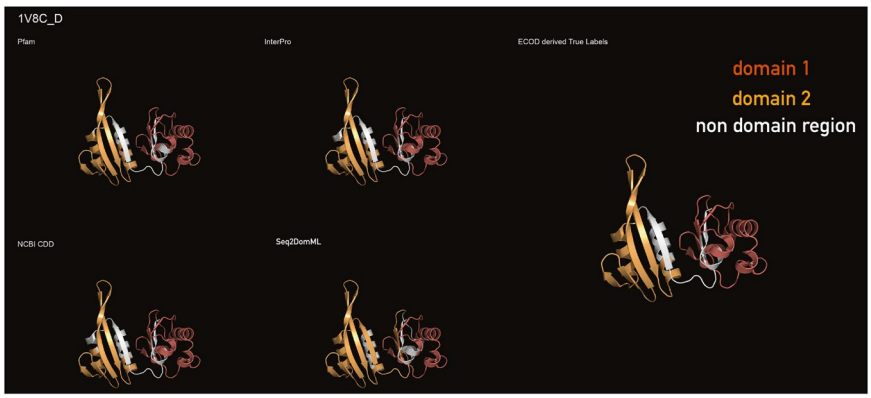


Fig. 7 Structural comparison of predicted domain partitions for PDB 1V8C chain D. Residue labels predicted by Pfam, InterPro, NCBI CDD, and Seq2DomML are mapped onto the experimentally determined protein structure and compared with the ECOD derived true labels. White regions represent residues assigned to the non-domain class, including terminal regions and boundaries separating domain units. Coloured regions represent residues assigned to the within-domain class. Distinct colours are used to separate consecutive predicted domain units. The ECOD structure on the right provides the reference domain partition against which the four prediction methods are visually compared.

Such cases should nevertheless be interpreted as hypothesis-generating observations rather than evidence that the ECOD assignment is incorrect.

For PDB structure 2YQB Chain A (Fig. 8), the ECOD-derived annotations contain three visually distinct domain regions. Pfam underdetects part of this structure, leaving a large central region white, which suggests that no domain was recognized across a substantial structured portion of the protein. InterPro recovers a multi-domain pattern that more closely resembles the ECOD-derived labels, although some boundary placements differ. NCBI CDD identifies the large right-hand domain but misses an important boundary, thereby merging two distinct domains. In contrast, Seq2DomML recovers a three-domain organization that, like the InterPro prediction, is visually similar to the ECOD segmentation. The left helical domain, central beta-rich domain, and larger right-hand beta-rich domain are all represented. Although some boundary regions extend into structured elements, this outcome is expected under the label-expansion strategy. For this structure, the Seq2DomML and InterPro segmentations appear to agree most closely with the ECOD reference.

PDB structure 5B7W Chain B (Fig. 9) provides a clear example in which existing tools fail to detect structured domain content. The ECOD-derived labels show two major colored regions separated by a white boundary-associated interval. Pfam and NCBI CDD assign the right-hand region as a single domain while leaving the folded left-hand region white, implying that no domain was detected across a structurally ordered portion of the protein. InterPro is even more conservative, leaving the entire structure white and therefore indicating no recognized domain. Seq2DomML, however, recovers a two-domain pattern that more closely resembles the ECOD-derived labels by assigning the left and right folded regions as separate domain units. It also retains the white linker between them, which could guide splice-site selection between independently folded units during construct design.

This example provides a strong visual illustration of the proposed model"s potential value as a domain-partitioning tool because it identifies structured domain content in regions not recognized by the other methods. Notably, this structure is annotated in the PDB as having an *unknown function*, supporting the possibility that undercharacterized domains may be more difficult for existing homology-based tools to recognize from sequence input. This example therefore illustrates the potential utility of Seq2DomML when domains are uncharacterized, unrecognized, or poorly represented in existing databases.

Overall, the structure-mapped comparisons show that the quantitative metrics only partially capture each tool"s capabilities of the prediction task. Pfam, InterPro, and NCBI CDD often perform well when the queried sequence matches existing domain models, but their visual outputs can contain long white regions where no domain is

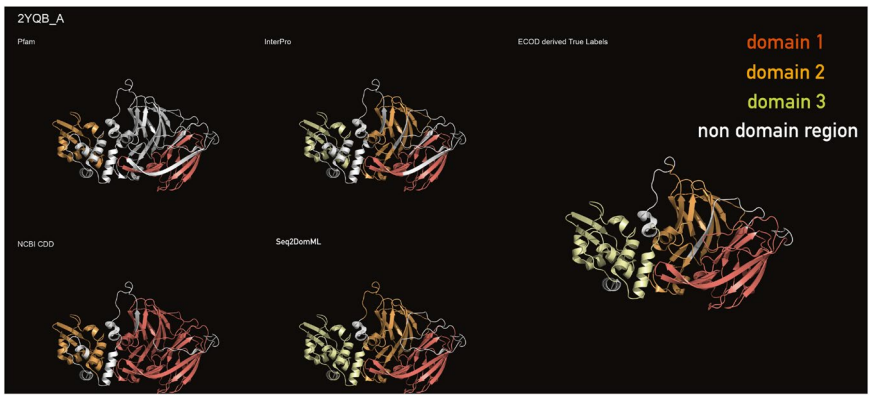


Fig. 8 Structural comparison of predicted domain partitions for PDB 2YQB chain A. Residue labels predicted by Pfam, InterPro, NCBI CDD, and Seq2DomML are mapped onto the experimentally determined protein structure and compared with the ECOD derived true labels. White regions represent residues assigned to the non-domain class, including terminal regions and boundaries separating domain units. Coloured regions represent residues assigned to the within-domain class. Consecutive domain units are shown using distinct colours. The ECOD structure on the right provides the reference partition for evaluating how closely each method reproduces the three-domain architecture of 2YQB Chain A.



Fig. 9 Structural comparison of predicted domain partitions for PDB 5B7W chain B. Residue labels predicted by Pfam, InterPro, NCBI CDD, and Seq2DomML are mapped onto the experimentally determined protein structure and compared with the ECOD derived true labels. White regions represent residues assigned to the non-domain class, including terminal regions and boundaries separating domain units, while coloured regions represent residues assigned to the within-domain class. Consecutive domain units are displayed using distinct colours. The ECOD structure on the right provides the reference partition for assessing how closely each method reproduces the two-domain architecture of 5B7W Chain B.

detected, even when those residues clearly participate in folded secondary structure. Conversely, Seq2DomML sometimes introduces additional domain partitions or boundary-associated regions absent in the true ECOD labels, but these predictions are frequently localized near regions that have debatable structural transitionary characteristics such as exposed loops, or separable folded lobes rather than being randomly distributed. This distinction is important for the intended application of the model. For construct design, a sequence-only predictor does not need to replace curated domain databases; rather, it can provide an additional residue-level hypothesis layer for identifying candidate splice regions, especially in cases where a domain is largely uncharacterized, leading to a complete absence of domain recognition.

Despite the contributions and potential impact of this study, it has some limitations. A direct experimental benchmark against attention-based architectures or pretrained protein language models was not conducted in the present study. The objective was to evaluate the proposed feature-engineered, task-specific framework, and incorporating these architectures would require substantially different input representations, pretraining resources, and model-development strategies. These differences would also make it more difficult to isolate the contribution of the sequence-derived feature-selection framework investigated here. Consequently, the reported results do not establish how Seq2DomML performs relative to modern transformer-based or pretrained protein language models. Direct benchmarking against these architectures therefore represents an important direction for future work.

The present evaluation was also limited to sequence-based domain annotation methods and did not include structure-based domain-parsing approaches applied to AlphaFold-predicted structures, such as DPAM. Such a comparison is technically feasible because AlphaFold models could be generated or retrieved for the same protein sequences and the resulting domain assignments could be mapped back to residue intervals. However, it would introduce predicted structural coordinates and a substantially different input and processing pipeline from the sequence-only setting investigated in this study. Therefore, the current results should be interpreted as a comparison among methods operating from amino acid sequence input rather than as a comprehensive comparison with structure-based domain-prediction pipelines. Evaluating Seq2DomML against AlphaFold-based domain-parsing approaches represents an important direction for future work.

Conclusion

Domain boundary prediction across protein sequences remains a crucial step in construct design for structural solution, particularly when selecting splice sites that improve the isolation, expression, and crystallization of stable single-domain protein units. In this study, ECOD-derived structural domain annotations were used to train a sequence-based bidirectional LSTM model capable of predicting residue-level domain and non-domain regions without requiring structural coordinates as input, producing the proposed framework of Seq2DomML. This approach was designed to support the identification of boundary, linker, and terminal flanking regions that may guide domain partitioning during construct design.

Established tools, including Pfam, InterPro, and NCBI CDD, use homology-based annotation to predict domain regions and therefore perform well when a domain has many well-studied homologs. However, they may fail to recognize domains when homologs are absent or poorly represented in their databases. Seq2DomML demonstrated competitive performance alongside these tools and showed potential advantages for poorly characterized domain families.

The model achieved the strongest macro F1-score, non-domain residue precision, non-domain residue F1-score, and domain IoU among the evaluated methods. In particular, the highest domain IoU indicates that Seq2DomML"s predicted within-domain regions showed the greatest spatial overlap with the ECOD-derived reference domain regions.

Overall, these results suggest that sequence-based learning from ECOD annotations can provide a complementary prediction layer for protein domain partitioning and structural solution workflows, particularly when homology-based tools fail to recognize structured regions with limited annotation coverage. Future work should focus on improving boundary and linker recall, experimentally validating predicted splice sites in expression and crystallization pipelines, and testing whether Seq2DomML generalizes beyond bacterial X-ray structures to broader structural biology applications.

Supplementary information The online version contains supplementary material available at https://doi.org/10.1186/s12859-026-06666-y.

Acknowledgements Not applicable.

Author contributions Author Alana Monks, contributed to the conceptualization of the study, methodological developments, software implementation, data collection, data preprocessing, model development, formal analysis and visualizations, as well as the preparation of this manuscript.Author Azam Asilian Bidgoli co-supervised this study, contributed to the to the conceptualization of the study and methodological developments, and the revision of the manuscript.Author Mike Suits conceived the study, defined the research problem, co-supervised the project, provided methodological guidance, contributed to funding acquisition, and revised the manuscript.All authors reviewed and approved the final manuscript.Query ID="Q1" Text="The Author Contributions statement appears here as it was entered in the submission system. However, we noticed that the wording was different in the manuscript file. Please check the statement and amend if necessary."

Funding Funding was granted by the Natural Sciences and Engineering Research Council of Canada (NSERC).

Data Availability The datasets and materials supporting the conclusions of this article are available in the Supplementary materials.The data made available includes the protein sequences retrieved from the PDB, their PDB chain identifiers, ECOD derived domain annotations, the full set of features, and the retained features after feature selection. Structural domain annotations utilized as the target variable were obtained from the October 2025 archive of the Evolutionary Classification of Domains database. Domain annotations used for comparison were obtained from Pfam, InterPro, and NCBI Conserved Domain Database.The relevant database versions, access dates, and accession information arereported in the supplementary materials repository. The public database used to collect protein sequences (PDB) can be accessed at https://www.rcsb.org/.Query ID="Q2" Text="The Data Availability statement appears here as it was entered in the submission system. However, we noticed that the wording was different in the manuscript file. Please check the statement and amend if necessary."

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Cooper CDO, Marsden BD. N- and c-terminal truncations to enhance protein solubility and crystallization: Predicting protein domain boundaries with bioinformatics tools. Methods Mol Biol. 2017;1586:11–31.
2. Edich M, Gao Y, Briggs DC, Thorn A. Unlocking the secrets of SARS-CoV-2 nsp3 by combining experiments with AlphaFold2 domain prediction. Life Sci. Alliance. 2026;9:e202503247.
3. Malawski GA, Hillig RC, Monteclaro F, Eberspaecher U, Schmitz AAP, Crusius K, et al. Identifying protein construct variants with increased crystallization propensity–a case study. Protein Sci. 2006;15:2718–28.
4. Paysan-Lafosse T, Andreeva A, Blum M, Chuguransky SR, Grego T, Pinto BL, et al. The pfam protein families database: embracing AI/ML. Nucleic Acids Res. 2025;53:D523–D534.
5. Blum M, Andreeva A, Florentino LC, Chuguransky SR, Grego T, Hobbs E, et al. InterPro: the protein sequence classification resource in 2025. Nucleic Acids Res. 2025;53:D444–D456.
6. Hart DJ, Waldo GS. Library methods for structural biology of challenging proteins and their complexes. Curr Opin Struct Biol. 2013;23:403–8.
7. Cheng H, Schaeffer RD, Liao Y, Kinch LN, Pei J, Shi S, et al. ECOD: an evolutionary classification of protein domains. PLoS Comput Biol. 2014;10:e1003926.
8. Burley SK, Bhatt R, Bhikadiya C, Bi C, Biester A, Biswas P, et al. Updated resources for exploring experimentally-determined PDB structures and computed structure models at the RCSB protein data bank. Nucleic Acids Res. 2025;53:D564–D574.
9. Kawashima S, Pokarowski P, Pokarowska M, Kolinski A, Katayama T, Kanehisa M. AAindex: amino acid index database, progress report 2008. Nucleic Acids Res. 2008;36:D202–D205.
10. Jaenicke R. Stability and folding of domain proteins. Prog Biophys Mol Biol. 1999;71:155–241.
11. Bhaskara RM, de Brevern AG, Srinivasan N. Understanding the role of domain-domain linkers in the spatial orientation of domains in multi-domain proteins. J Biomol Struct Dyn. 2013;31:1467–80.
12. Holcomb J, Spellmon N, Zhang Y, Doughan M, Li C, Yang Z. Protein crystallization: Eluding the bottleneck of x-ray crystallography. AIMS Biophys. 2017;4:557–75.
13. Reddy Chichili VP, Kumar V, Sivaraman J. Linkers in the structural biology of protein-protein interactions. Protein Sci. 2013;22:153–67.
14. Papaleo E, Saladino G, Lambrughi M, Lindorff-Larsen K, Gervasio FL, Nussinov R. The role of protein loops and linkers in conformational dynamics and allostery. Chem Rev. 2016;116:6391–423.
15. Ito S, Nishino T. Rational engineering in protein crystallization: Integrating physicochemical principles, molecular scaffolds, and computational design. Crystals (Basel). 2025;16:36.
16. Bhaskara RM, Srinivasan N. Stability of domain structures in multi-domain proteins. Sci Rep. 2011;1:40.
17. Hochreiter S, Schmidhuber J. Long short-term memory. Neural Comput. 1997;9:1735–80.
18. Li S, Chen J, Liu B. Protein remote homology detection based on bidirectional long short-term memory. BMC Bioinformatics. 2017;18:443.
19. Boadu F, Lee A, Cheng J. Deep learning methods for protein function prediction. Proteomics. 2025;25:e2300471.
20. Xuan W, Liu N, Huang N, Li Y, Wang J. CLPred: a sequence-based protein crystallization predictor using BLSTM neural network. Bioinformatics. 2020;36:i709–i717.
21. Liu J, Tang X, Guan X. Grain protein function prediction based on self-attention mechanism and bidirectional LSTM. Brief Bioinform. 2023;24:bbac493.

22. Gong Y, Li R, Fu B, Liu Y, Wang J, Li R, et al. A CNN-LSTM ensemble model for predicting protein-protein interaction binding sites. IEEE/ACM Trans Comput Biol Bioinform. 2023;20:3588–99.
23. Wang L, Zhong H, Xue Z, Wang Y. Res-Dom: predicting protein domain boundary from sequence using deep residual network and Bi-LSTM. Bioinform. Adv. 2022;2:vbac060.
24. Zhang Y, Bian B, Okumura M. Hyena architecture enables fast and efficient protein language modeling. IMetaOmics. 2025;2:e45.
25. Coello Coello CA, Lamont GB, van Veldhuizen DA. Evolutionary algorithms for solving multi-objective problems. Genetic and Evolutionary Computation. 2nd Edition. New York, NY: Springer; 2007.
26. Wang Y, Fan R, Cheng L, Gong B, Liu J. An improved adaptive NSGA-II with multiple filtering for high-dimensional feature selection. Electronics (Basel). 2026;15:236.
27. Deb K, Jain H. An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part i: Solving problems with box constraints. IEEE Trans Evol Comput. 2014;18:577–601.
28. Wang J, Chitsaz F, Derbyshire MK, Gonzales NR, Gwadz M, Lu S, et al. The conserved domain database in 2023. Nucleic Acids Res. 2023;51:D384–D388.
29. Varadi M, Bertoni D, Magana P, Paramval U, Pidruchna I, Radhakrishnan M, et al. AlphaFold protein structure database in 2024: providing structure coverage for over 214 million protein sequences. Nucleic Acids Res. 2024;52:D368–D375.
30. The UniProt Consortium. UniProt: the universal protein knowledgebase in 2025. Nucleic Acids Res. 2025;53:D609–D617.
31. Letunic I, Bork P. 20 years of the SMART protein domain annotation resource. Nucleic Acids Res. 2018;46:D493–D496.
32. Mi H, Lazareva-Ulitsky B, Loo R, Kejariwal A, Vandergriff J, Rabkin S, et al. PANTHER database of protein families, subfamilies, functions and pathways. Nucleic Acids Res. 2005;33:D284–D288.
33. Hulo N, Bairoch A, Bulliard V, Cerutti L, De Castro E, Langendijk-Genevaux PS, et al. The PROSITE database. Nucleic Acids Res. 2006;34:D227–D230.
34. Durham J, Zhang J, Schaeffer RD, Cong Q. DPAM-AI: a domain parser for AlphaFold models powered by artificial intelligence. Bioinformatics. 2024;41:btae740.
35. Schaeffer RD, Zhang J, Medvedev KE, Kinch LN, Cong Q, Grishin NV. ECOD domain classification of 48 whole proteomes from AlphaFold structure database using DPAM2. PLoS Comput Biol. 2024;20:e1011586.
36. Triant DA, Pearson WR. Most partial domains in proteins are alignment and annotation artifacts. Genome Biol. 2015;16:99.
37. Nogales E, Louder RK, He Y. Cryo-EM in the study of challenging systems: the human transcription pre-initiation complex. Curr Opin Struct Biol. 2016;40:120–7.
38. Sharma S, Zheng H, Huang YJ, Ertekin A, Hamuro Y, Rossi P, et al. Construct optimization for protein NMR structure analysis using amide hydrogen/deuterium exchange mass spectrometry. Proteins. 2009;76:882–94.
39. Kermani AA. A guide to membrane protein x-ray crystallography. FEBS J. 2021;288:5788–804.
40. Grabowicz M. Lipoproteins and their trafficking to the outer membrane. EcoSal Plus. 2019;8:67–76.
41. A Bushuiev, R Bushuiev, J Sedlar, T Pluskal, J Damborsky, S Mazurenko, et al., Revealing data leakage in protein interaction benchmarks (Apr. 2024). arXiv:2404.10457 [Preprint] https://arxiv.org/abs/2404.10457.Query ID="Q4" Text="If ref. 41 (preprint) has now been published in final peer-reviewed form, please update the reference details if appropriate."
42. Lertampaiporn S, Nuannimnoi S, Vorapreeda T, Chokesajjawatee N, Visessanguan W, Thammarongtham C. PSO-LocBact: A consensus method for optimizing multiple classifier results for predicting the subcellular localization of bacterial proteins. Biomed Res Int. 2019;2019:5617153.
43. Kawashima S, Kanehisa M. Aaindex: Amino acid index database. Nucleic Acids Res. 2000;28:374.
44. Y Cui, M Jia, T-Y Lin, Y Song, S Belongie, Class-balanced loss based on effective number of samples, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, 9260-9269.
45. Chen B, Bao W. OPMD_TCM_PTM: a post-translational modification sites identification model for oral potentially malignant diseases" traditional chinese medicine proteins. Biomed Signal Process Control. 2026;125:110717.
46. Bao W, Yang B, Chen B. TAPE_selection: Organelle proteins classification with TAPE feature selection. IEEE/ACM Transactions on Computational Biology and Bioinformatics. 2025;22:1542–52.
47. Asilian Bidgoli A, Rahnamayan S. Multi-phase evolutionary search space shrinking for large-scale multi-objective feature selection. Appl. Soft Comput. 2025;184:113755.
48. Hou Q, Waury K, Gogishvili D, Feenstra KA. Ten quick tips for sequence-based prediction of protein properties using machine learning. PLoS Comput Biol. 2022;18:e1010669.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Alana Monks[1] · Azam Asilian Bidgoli[2] · Michael D. L. Suits[1]

✉ Azam Asilian Bidgoli
abidgoli@wlu.ca

✉ Michael D. L. Suits
msuits@wlu.ca

[1] Chemistry and Biochemistry, Wilfrid Laurier University, 75 University Avenue West, Waterloo N2L3C5, Ontario, Canada

[2] Computer Science and Physics, Wilfrid Laurier University, 75 University Avenue West, Waterloo N2L3C5, Ontario, Canada